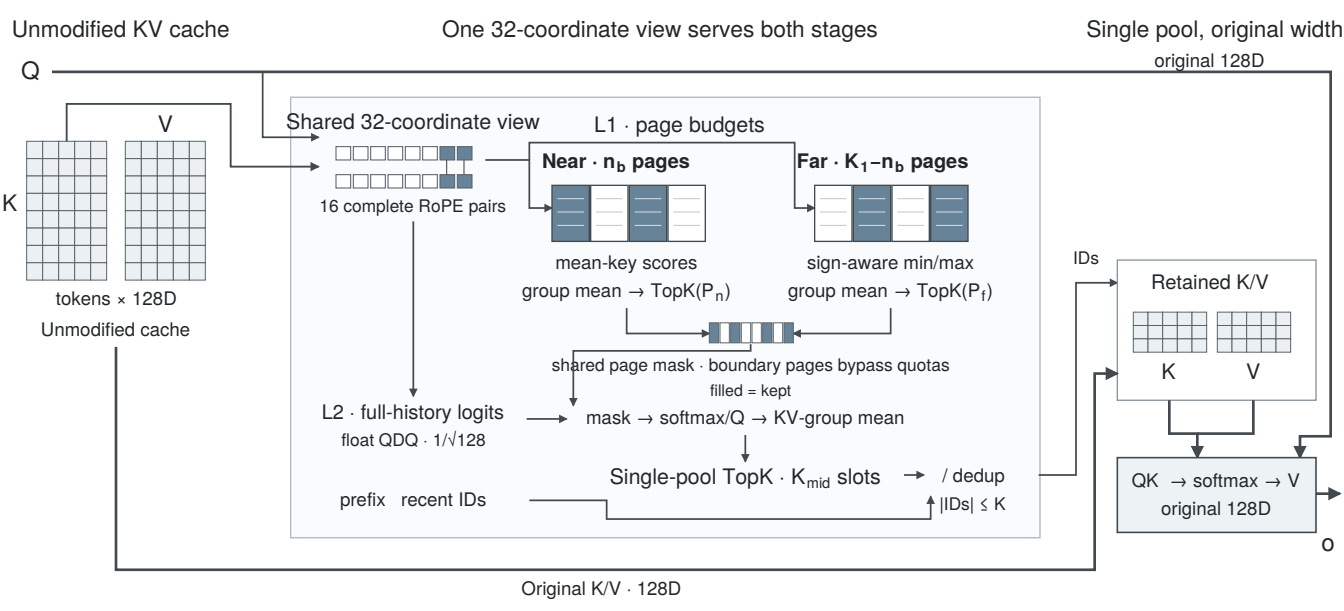
Unmodified KV cache
One 32-coordinate view serves both stages
Single pool, original width
Q
original 128D
V
K
tokens × 128D
Unmodified cache
Shared 32-coordinate view
16 complete RoPE pairs
L1 · page budgets
Near · n_b pages
Far · $K_1 - n_b$ pages
mean-key scores
group mean → TopK(P_n)
sign-aware min/max
group mean → TopK(P_f)
shared page mask · boundary pages bypass quotas
filled = kept
L2 · full-history logits
float QDQ · 1/√128
mask → softmax/Q → KV-group mean
Single-pool TopK · K_{mid} slots
/ dedup
prefix recent IDs
|IDs| ≤ K
IDs
Retained K/V
K
V
QK → softmax → V
original 128D
o
Original K/V · 128D
Optional far gate: prefill-calibrated, frozen at decode · disabled in this reference